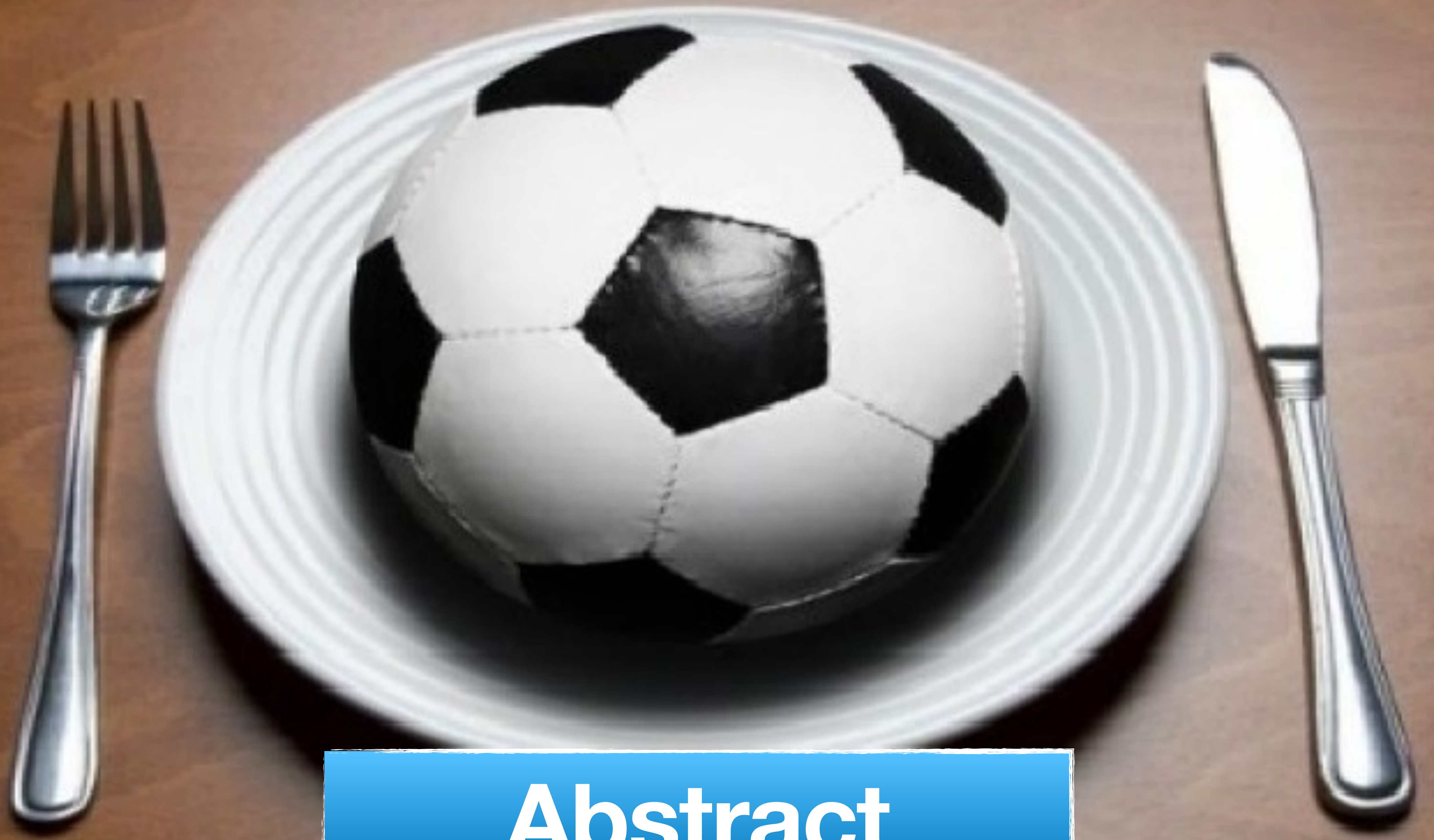


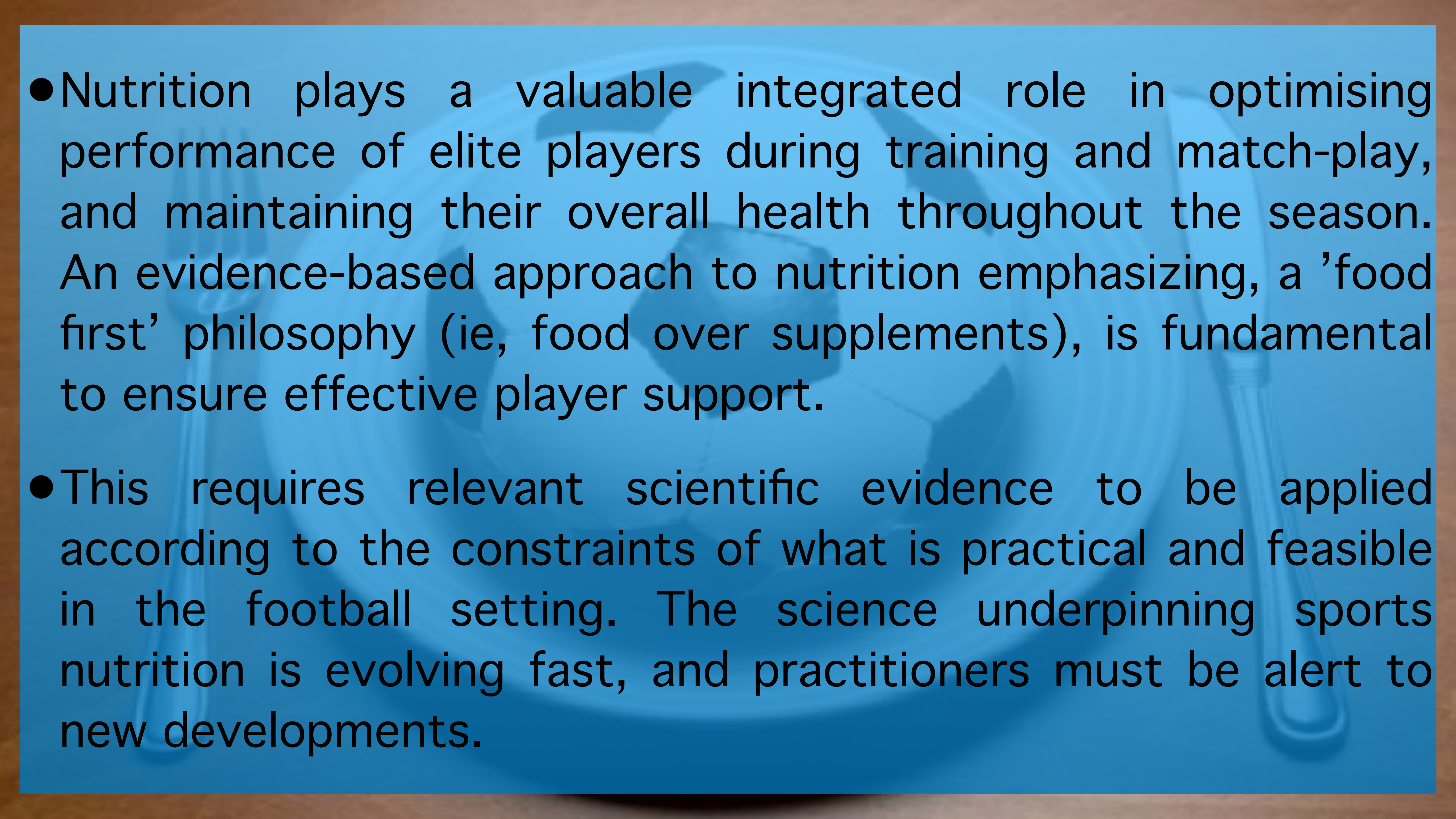
MATCH DAY NUTRITION



Collins, J. et al. UEFA expert group statement on nutrition in elite football Current evidence to inform practical recommendations and guide future research. (Br J Sports Med, 2020)

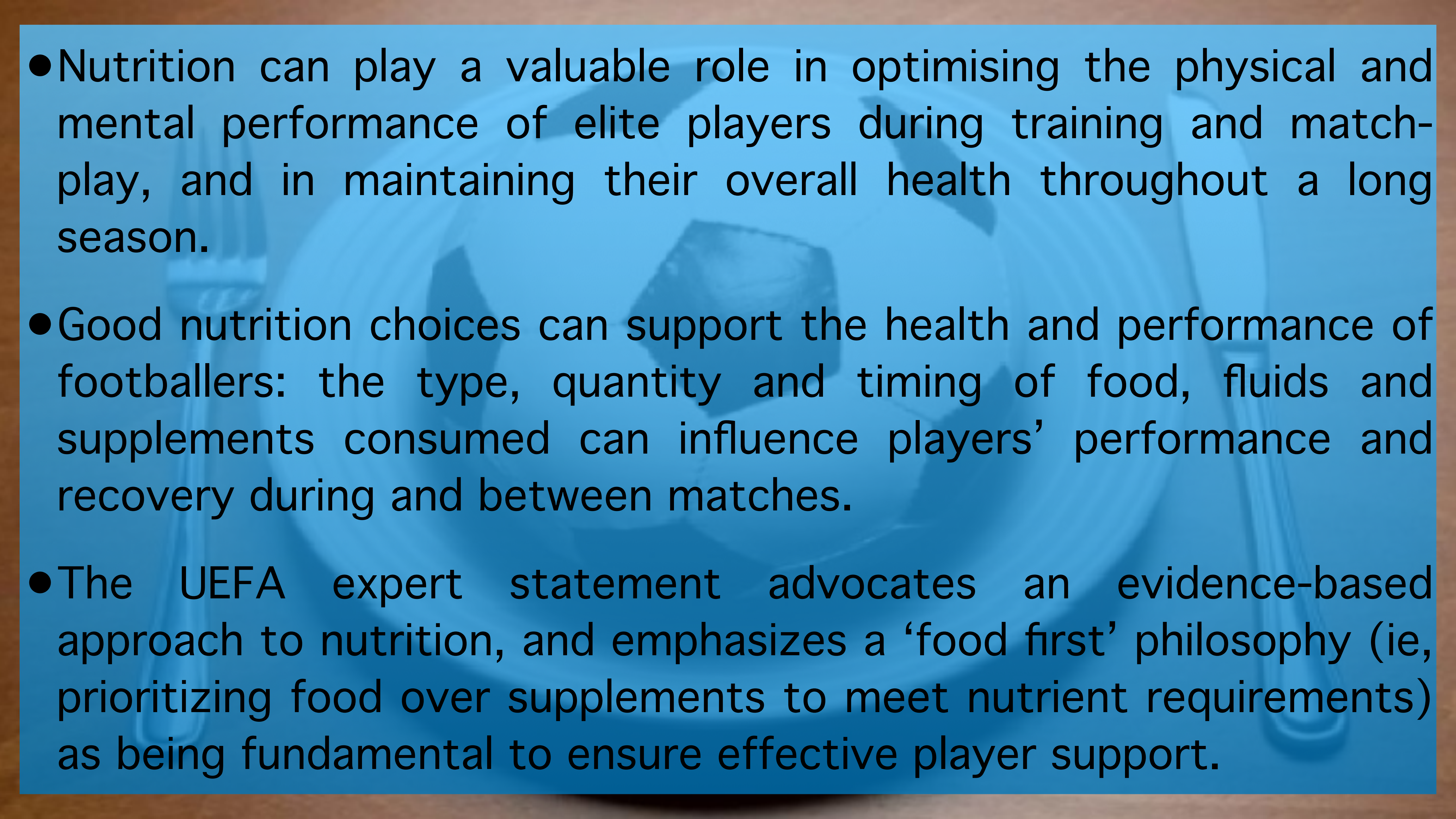


Abstract

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- Nutrition plays a valuable integrated role in optimising performance of elite players during training and match-play, and maintaining their overall health throughout the season. An evidence-based approach to nutrition emphasizing, a 'food first' philosophy (ie, food over supplements), is fundamental to ensure effective player support.
 - This requires relevant scientific evidence to be applied according to the constraints of what is practical and feasible in the football setting. The science underpinning sports nutrition is evolving fast, and practitioners must be alert to new developments.

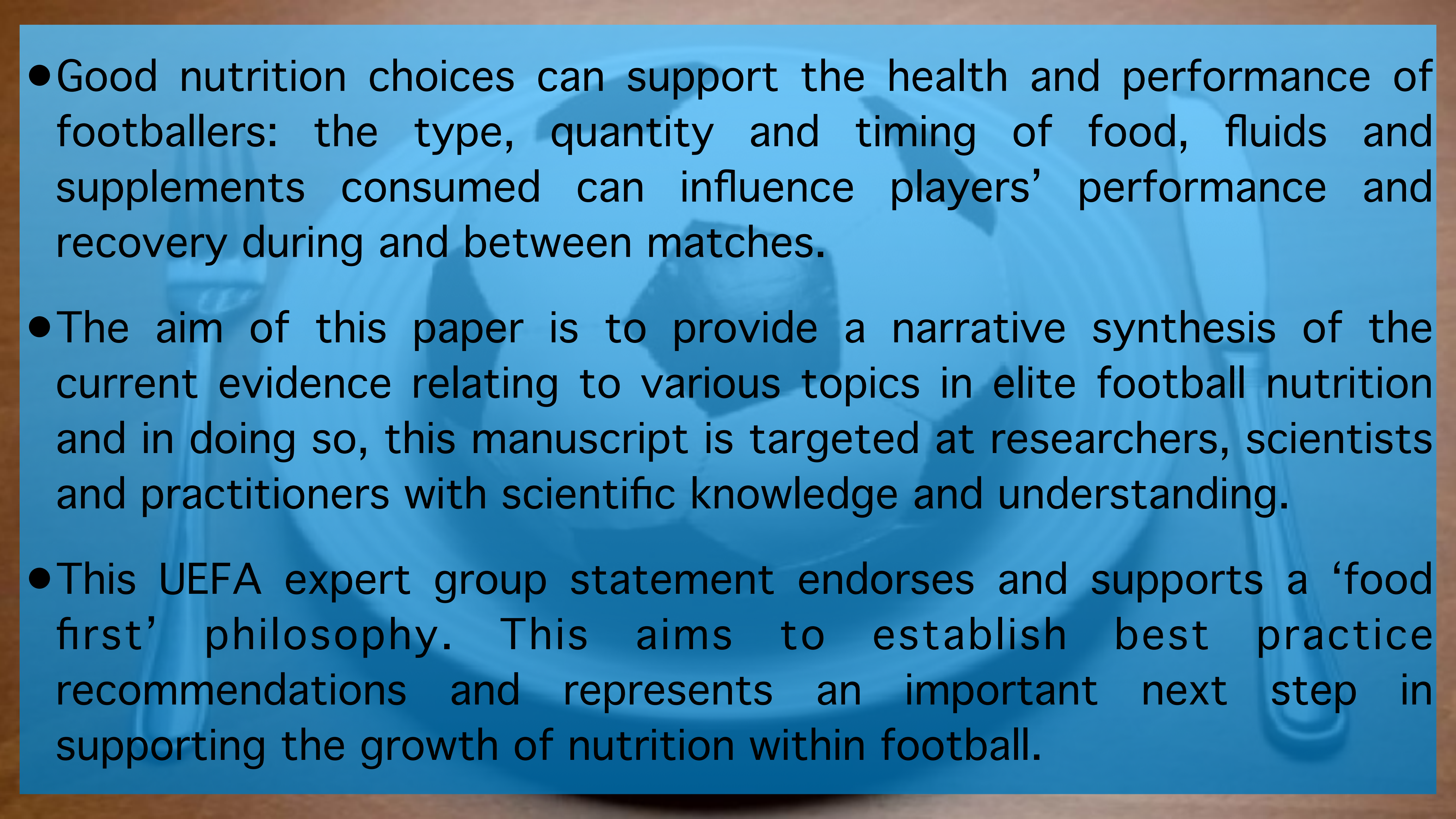


Executive Summary

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- Nutrition can play a valuable role in optimising the physical and mental performance of elite players during training and match-play, and in maintaining their overall health throughout a long season.
 - Good nutrition choices can support the health and performance of footballers: the type, quantity and timing of food, fluids and supplements consumed can influence players' performance and recovery during and between matches.
 - The UEFA expert statement advocates an evidence-based approach to nutrition, and emphasizes a 'food first' philosophy (ie, prioritizing food over supplements to meet nutrient requirements) as being fundamental to ensure effective player support.



Introduction

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- Good nutrition choices can support the health and performance of footballers: the type, quantity and timing of food, fluids and supplements consumed can influence players' performance and recovery during and between matches.
 - The aim of this paper is to provide a narrative synthesis of the current evidence relating to various topics in elite football nutrition and in doing so, this manuscript is targeted at researchers, scientists and practitioners with scientific knowledge and understanding.
 - This UEFA expert group statement endorses and supports a 'food first' philosophy. This aims to establish best practice recommendations and represents an important next step in supporting the growth of nutrition within football.



Match Day Nutrition

Preparation for match play (carbohydrates and fluids)

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- CHO is the primary fuel for muscle during high-intensity activities; it is therefore a key macronutrient when preparing players for match play. On the day prior to a match (MD-1), training is usually light and CHO intake should be at least 6–8g/kg body mass (BM) to elevate muscle and liver glycogen stores.
- Players who begin a game with low muscle glycogen stores will cover less distance and much less at high speed, particularly in the second half, than those who have ensured adequate glycogen stores.

Preparation for match play (carbohydrates and fluids)

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- Where the match schedule consists of congested fixtures (eg, domestic fixtures, European competition, international games), CHO intake should be maintained within this range (6–8g/kg BM/day) for the 48–72hours between games to promote adequate glycogen storage.
- A conscious focus on the intake of CHO-rich foods is needed, with increased CHO intake at the cost of fat intake (and possibly protein intake) to ensure adequate glycogen restoration.

Preparation for match play (carbohydrates and fluids)

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- Maintaining an appropriate hydration status will support players' health and performance.
- Sweating is the primary mechanism to dissipate the metabolic heat generated during football training and match play in both cool and hot environments.
- Players should aim to start the match fully hydrated.
- For games with an early kick-off, the day prior to the match represents a key opportunity to optimise the players' hydration status for the match the following day.

Pre-match (carbohydrates and fluids)

- On MD itself, CHO intake is again one of the most important considerations. Within an overall guideline of 6–8g/kg BM CHO per day, it is recommended that players consume a CHO-rich meal (1–3g/kg BM) 3–4hours before kick-off to ensure that they begin the match with adequate glycogen stores.
- The pre-match meal is of particular importance for the promotion of liver glycogen stores, given that such stores can be reduced by about 50% after an overnight fast.
- This may be particularly important for matches with a lunchtime kick-off, and it highlights the importance of optimising nutritional preparation during the day prior to the match. The pre-match meal should be easily digestible to reduce the risk of gastrointestinal problems (eg, reflux, discomfort). The pre-match meal should also make players ‘feel better’ so comfort should be considered, rather than rigid strategies focused solely on meeting CHO intake guidelines.
- Player ‘rituals’ can be strongly held and education combined with practicing pre-match fueling in training or lower priority matches, can be an important tool to optimise glycogen stores and player readiness for match play.

Pre-match (carbohydrates and fluids)

- Data from many studies suggest that high CHO intakes before and during a match can delay fatigue and enhance the capacity for intermittent high-intensity exercise.
- Benefits of pre-match meals may extend to players' technical performance. For example, increased dribbling speed was observed when professional youth footballers consumed a larger breakfast (500 vs 250 kcal, with 60% CHO) 135 min before a match.
- Finally, players should aim to start the match euhydrated by ingesting 5–7mL/kg BM of fluid in the 2–4hours prior to kick-off. This allows time for excess fluid to be voided prior to exercise, targeting a urine that is pale yellow in colour.

During match play (carbohydrates and fluids)

- Sufficient CHO and fluid intake are the two main nutritional considerations during match play.
- Research evidence typically shows performance benefits in protocols simulating football matches when CHO is consumed during exercise at rates of ~30–60g/hour, or when 60g is consumed before each half.
- It is therefore recommended that ~30–60g CHO is consumed after warm-up and again at half-time to meet these guidelines.
- Sports foods (eg, CHO drinks gels) can provide a preferred delivery option, to minimize gastrointestinal issues.
- Stoppages during the match may also provide a valuable opportunity for players with increased CHO and/or fluid needs.

During match play (carbohydrates and fluids)

- Receptors in the oral cavity detect CHO consumed during exercise and exert central effects that may reduce the perception of effort.
- CHO mouth rinsing has been shown to increase self-selected jogging speed with likely benefits in sprint performance during intermittent exercise.
- The use of CHO mouth rinsing during breaks in match play (eg, half-time, extra-time, injury stoppages, medical breaks) could potentially enhance performance in situations where CHO consumption is limited by gastrointestinal concerns.
- Swallowing the drink following a ~5 s exposure in the mouth could be beneficial in matches that can extend to extra time and penalty shoot-outs where both the brain and muscle may benefit from additional fuel support and activation.

During match play (carbohydrates and fluids)

- It is likely that individual players may be more or less sensitive to hypohydration during exercise. Therefore as a guide, players should aim to drink sufficient fluids to prevent a deficit of $>2\%$ – 3% of pre-exercise BM during exercise, while avoiding gains in BM (hyperhydration) and also ensuring their fuel needs are met.
- Both hydration and CHO intake may require special attention in matches where extra time ($2 \times 15\text{min}$) is played. All match nutrition strategies, including the use of supplements (eg, sports foods), should be practised in training and minor matches to allow individualized protocols to be developed and to identify adverse effects in players, as well as allowing them the opportunity to become accustomed to any potential adverse effects, with minimal impact on important match performance.

Recovery from match play (CHO, fluids, protein)

- A primary objective following a competitive match is to reduce the time needed to fully recover.
- One essential goal is to rapidly replenish CHO stores. Postmatch meals and snacks should target a CHO intake of $\sim 1 \text{ g/kg BM/hour}$ for 4 hours. This is usually facilitated by the consumption of drinks and snacks in the changing rooms followed by post-match meals at the stadium, during travel and at home.
- Sports foods may provide a preferred option to supply macronutrients, especially to achieve CHO guidelines when appetite may be reduced or when sourcing food away from the home environment.

Recovery from match play (CHO, fluids, protein)

- Players should also aim to reduce any fluid and electrolyte deficit soon after the match; however, in most situations there is sufficient opportunity and time to restore euhydration and electrolyte balance with normal eating/drinking practices, while also meeting other recovery objectives.
- Elite football players may not achieve CHO targets in recovery from evening games, suggesting suboptimal glycogen resynthesis patterns, the result of which is likely problematic for recovery and preparation during congested fixture schedules.
- Daily CHO intake in the range of 6–8g/kg BM in the 24 hours following a game (MD +1) will continue to replenish glycogen stores and this intake should be maintained for up to 48–72 hours after the match during congested fixture schedules.
- Higher intakes and additional nutritional strategies may be required when players report symptoms of muscle soreness and damage, as glycogen synthesis is impaired in the presence of muscle damage.

Recovery from match play (CHO, fluids, protein)

- To optimise protein synthesis for repair and adaptation, meals and snacks should be scheduled to achieve intakes of 20–25g of (high-quality) protein at 3–4hour intervals.
- Furthermore, there is emerging evidence that consuming 30–60g of casein protein prior to sleep can enhance overnight protein synthesis.
- Reducing exercise-induced muscle inflammation and free radical production, particularly with large doses of individual antioxidant vitamins C and E, may interfere with adaptive processes in muscle and is therefore discouraged.